Hyperbolic Functions: Inverse Functions

Questions 1–33

Question 1. For each of the following functions, sketch the graph:

a) $y = \operatorname{arsinh} x$

b) $y = \operatorname{arcosh} x$

c) $y = \operatorname{artanh} x$

Question 2. Given that $x \in \mathbb{R}$, show that

$$\operatorname{arsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right).$$

Question 3. Given that $x \in \mathbb{R}$ and $k > 0$, show that

$$\operatorname{arsinh} \frac{x}{k} = \ln \left(\frac{x + \sqrt{x^2 + k^2}}{k} \right).$$

Question 4. Given that $x \geq 1$, show that

$$\operatorname{arcosh} x = \ln \left(x + \sqrt{x^2 - 1} \right).$$

Question 5. Given that $\frac{x}{k} \geq 1$ and $k > 0$, show that

$$\operatorname{arcosh} \frac{x}{k} = \ln\left(\frac{x + \sqrt{x^2 - k^2}}{k}\right).$$

Question 6. Given that $-1 < x < 1$, show that

$$\operatorname{artanh} x = \frac{1}{2} \ln\left(\frac{1 + x}{1 - x}\right).$$

Question 7. Given that $-k < x < k$ and $k > 0$, show that

$$\operatorname{artanh} \frac{x}{k} = \frac{1}{2} \ln \left(\frac{k+x}{k-x} \right).$$

Question 8. Solve the equation $\operatorname{arcosh} x = \ln 10$, writing your answer in exact form.

Question 9. Solve the equation $\operatorname{arsinh} x = -\ln 6$, writing your answer in exact form.

Question 10. Solve the equation $\operatorname{artanh} \frac{x}{6} = -\ln 2$, writing your answer in exact form.

Question 11. Solve the following equations, writing your answers in exact form:

a) $\operatorname{artanh} x = \ln\left(\sqrt{x+2}\right)$

b) $\operatorname{artanh} \frac{x}{3} = \ln\left(\sqrt{3-2x}\right)$

c) $\operatorname{artanh} 5x = \ln\left(\sqrt{2+5x}\right)$

d) $4x + 1 = \tanh(\ln(\sqrt{5-2x}))$

Question 12. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arsinh} x$

b) $y = \operatorname{arcosh} x$

c) $y = \operatorname{artanh} x$

Question 13. State the exact range of the following functions, given $-\frac{1}{2} < x < \frac{1}{2}$:

a) $f(x) = \operatorname{arsinh} x$

b) $g(x) = \operatorname{artanh} x$

Question 14. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arcosh} x + 1$

b) $y = \operatorname{arcosh}(x - 1)$

c) $y = \operatorname{arcosh} 2x$

d) $y = 2 \operatorname{arcosh} x$

e) $y = 3 \operatorname{arcosh}(x + 2)$

f) $y = \operatorname{arcosh}(2x - 3)$

g) $y = 4 \operatorname{arcosh} x - 5$

h) $y = \operatorname{arcosh}(4 - x)$

i) $y = 8 - 3 \operatorname{arcosh} x$

j) $y = -\operatorname{arcosh}\left(9 - \frac{1}{2}x\right)$

Question 15. State the largest possible domain and range of the following functions:

a) $y = \operatorname{arsinh}^2 x$

b) $y = \operatorname{arcosh}^2 x$

c) $y = \operatorname{artanh}^2 x$

Question 16. Given that $x \geq 1$, explain why $\operatorname{arsinh} x > \operatorname{arcosh} x$.

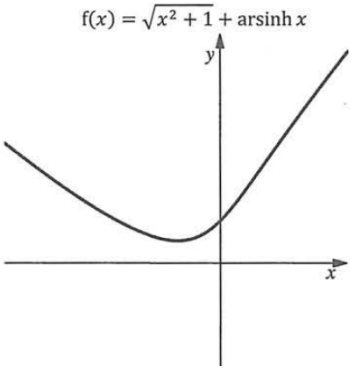
Question 17. Given that $x \geq 1$, state the exact range of $f(x) = \operatorname{arsinh} x + \operatorname{arcosh} x$.

Question 18. Solve the equation

$$\operatorname{artanh} x + \ln \sqrt{1-x} = \frac{1}{2} \ln 3 - 2 \ln 2.$$

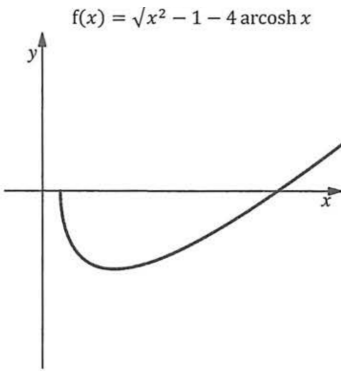
Question 19. Solve the equation $\operatorname{arsinh} x = \ln 4 - \ln 3 + \operatorname{arcosh} x$.

Question 20. The graph below shows $f(x) = \sqrt{x^2+1} + \operatorname{arsinh} x$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 21. The graph below shows $f(x) = \sqrt{x^2 - 1} - 4\,\text{arcosh}\,x$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 22. It is given that $f(x) = 2\,\text{arsinh}\,x - \text{arcosh}\,x$.

a) Explain why the largest possible domain of f is $\{x : x \in \mathbb{R},\ x \geq 1\}$.

b) Show that $y = f(x)$ has a single stationary point.

c) Show that the range of f is greater than or equal to

$$2\ln(\sqrt{15} + 2\sqrt{6}) - \ln(\sqrt{15} + \sqrt{6}) - \ln 3.$$

Question 23. Given that $x \neq 0$, show that

$$\operatorname{arcosech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1}\right).$$

Question 24. Given that $x \neq 0$ and $k > 0$, show that

$$\operatorname{arcosech} \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 + x^2}}{x}\right).$$

Question 25. Given that $0 < x < 1$, show that

$$\operatorname{arsech} x = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2} - 1}\right).$$

Question 26. Given that $0 < \frac{x}{k} < 1$ and $k > 0$, show that

$$\operatorname{arsech} \frac{x}{k} = \ln\left(\frac{k + \sqrt{k^2 - x^2}}{x}\right).$$

Question 27. Given that $x < -1$ or $x > 1$, show that

$$\operatorname{arcoth} x = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right).$$

Question 28. Given that $x < -1$ or $x > 1$, and $k > 0$, show that

$$\operatorname{arcoth} \frac{x}{k} = \frac{1}{2} \ln \left(\frac{x+k}{x-k} \right).$$

Question 29. Solve the equation $\operatorname{arcosech} x = \ln 3$, writing your answer in exact form.

Question 30. Solve the equation $\operatorname{arsech} \frac{x}{4} = \ln 5$, writing your answer in exact form.

Question 31. Solve the equation $\operatorname{arcoth} x = \ln 2$, writing your answer in exact form.

Question 32. Solve the following equations, writing your answers in exact form:

a) $\operatorname{arcosech} x = \ln\left(2 + \frac{1}{x}\right)$

b) $\operatorname{arsech} \frac{x}{2} = \ln\left(3 - \frac{4}{x}\right)$

c) $\operatorname{arcoth} 3x = \ln\sqrt{3 + 4x}$

Question 33. Solve the equation $\operatorname{artanh} \frac{x}{2} = \operatorname{arcoth}(3x + 1)$.